

Keyuan Hu

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EDUCATION

University of Michigan

B.S. in Mathematics (Honors)

Ann Arbor, MI

May 2026

- **GPA:** 4.0/4.0
- **Course Highlights:** Distributed Systems, ML, Combinatorics & Graph Theory, Measure Theory

Shanghai Jiao Tong University

B.S.E. in Electrical and Computer Engineering

Shanghai

August 2026

- **GPA:** 3.77/4.0
- **Course Highlights:** Analog/Digital Circuits, Computer Architecture
- **Awards/Honors:** Two terms Outstanding Undergraduate Scholarship (awarded to top 5% of college), Tang Junyuan Scholarship (awarded to only 5 students in the college)

Research Experience

Human-AI Lab

Research Assistant

Ann Arbor, MI

May. 2025-Sept. 2025

- Developed a full multimodal gaze-tracking and ML calibration pipeline, achieving an 18–25% reduction in average gaze error in simulated surgical environments.
- Collected and curated gaze–video datasets from 50 participants in OR-simulation tasks and contributed analyses, figures, and writing to an HCI conference manuscript.
- Built both an interactive research visualization platform and a project outreach website to present gaze overlays, timelines, and study impact to collaborators and external audiences.

Shanghai Jiao Tong University

Research Team Member

Shanghai

May. 2024-Oct. 2024

- Implemented a microorganism segmentation pipeline in ImageJ/Fiji using the StarDist star-convex model, significantly improving *E. coli* identification in denoised microscopy images
- Contributed to building a semi-real-time 3D tracking system that integrates segmentation outputs for stable long-term microorganism observation on real microscope setups.

Work Experience

Grandlynn Information Technologies Co., Ltd.

Intern

Su Zhou

Dec. 2023-Jan. 2024

- Supported backend maintenance for city-wide municipal monitoring platforms (shared-bike management, urban lighting), working with Java services and Maven-based builds.
- Assisted debugging, API updates, and database adjustments in production environments, gaining experience with enterprise software workflows.
- Participated in weekly requirement meetings with municipal clients and project managers to clarify feature needs, negotiate scope, and coordinate development timelines, strengthening professional communication skills.

Shanghai Jiao Tong University

Teaching Assistant

Shanghai

Aug. 2023-July. 2024

- Served as a full-time TA for the whole engineering honors math sequence (Math 186/285/286), delivering recitation sessions, holding office hours, designing practice problems, and assisting with grading.

Publication

SurgGaze: Implicit Calibration for Accurate Gaze Analysis in Operating Rooms with Wearable Eyetrackers

Wang, J., Natalie, R., **Hu, K.**, Xu, W., George, B., Popov, V., Guo, A., & Wang, X.

Under Review at CHI 2026